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Submission Summary

Conference Name
IEEE International Conference on Smart Power, Energy, Renewables, and Transportation (IEEE-SPERT)

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T05: Energy Storage and Management Technologies

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113

Paper Title
A Fail-Safe String-Level Reconfigurable Battery Management System Using Discrete Back-to-Back P-MOSFET Switches Design and Rover Validation on a 3S2P Li-Ion Pack

Abstract
Lithium-ion strings that are hard-wired in parallel cannot isolate one faulted string without disconnecting the whole pack. This paper presents the design, firmware and hardware validation of a string-level reconfigurable battery management system (BMS) for a 3S2P Samsung INR18650-30Q pack that powers a tracked mobile rover. Each string, and the load, is switched by a bidirectional high-side channel built from a common-source back-to-back IRF4905 P-channel MOSFET pair and a discrete NPN level shifter; base pull-down resistors hold every channel open whenever the microcontroller is in reset or unpowered. Firmware on an STM32F446RE runs a deterministic 20 ms loop in which a latching safety layer may only open switches, and a separate reconfiguration state machine is the only code permitted to close one. A close is allowed only after an inter-string voltage gate that bounds the true difference below 100 mV under load, break-before-make confirmed by current readback, and a 500 ms cooldown; any state the controller cannot explain forces a reset-only lockout. The unmodified safety and reconfiguration

modules passed 22 and 18 host-simulated scenarios and 800 randomized runs without an invariant violation. On hardware, the level shifter produced a gate-source voltage of −10.8 V, both sensors reported string voltages with zero I²C errors across all switching states, and on the rover an injected over-current isolated the affected string while the other string kept the rover driving in 20 of 20 trials for each string. The absence of per-cell sensing, temperature sensing and a precharge path is stated as a limitation.

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Primary Subject Area
2. Battery Management Systems

Secondary Subject Areas
3. State-of-Charge and Health Estimation

Submission Files
A Fail-Safe String-Level Reconfigurable Battery Management System Using Discrete Back-to-Back P-MOSFET Switches Design and Rover Validation on a 3S2P Li-Ion Pack.pdf (559.1 Kb, 2026-11-02 09:41:28 PST)